

Commissioned Review: TreS–Mak–GlgE Branched α -Glucan Biosynthesis in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Target bucket:** KEGG ppu00500 "Starch and sucrose metabolism" **Module under review:** TreS–Mak–GlgE–GlgB branched α -glucan biosynthesis (generic reusable bacterial module)

1. Executive Summary

The four-step **TreS–Mak–GlgE–GlgB** module that channels trehalose into intracellular branched α -glucan is **fully satisfiable** in *Pseudomonas putida* KT2440. Every reaction role maps cleanly onto a candidate gene within a single contiguous glycogen/trehalose locus spanning **PP_4050–PP_4060**. The defining architectural feature in KT2440 is that **steps 1 (trehalose synthase, TreS) and 2 (maltokinase, Mak) are fused into one bifunctional 1106-aa polypeptide, PP_4059 (treSB, Q88FN0)**, carrying two EC numbers (5.4.99.16 and 2.7.1.175). Steps 3 and 4 are encoded by dedicated single-copy genes: **GlgE = PP_4060 (Q88FM9, EC 2.4.99.16)** and **GlgB = PP_4058 (Q88FN1, EC 2.4.1.18)**. In mycobacteria, by contrast, TreS and the maltokinase Pep2 are *separate* proteins that assemble into a hetero-octameric complex — so KT2440 represents an architectural variant of the same two reaction roles, not a functional difference.

Critically, KT2440 **lacks GlgC (ADP-glucose pyrophosphorylase, EC 2.7.7.27) and the OtsAB trehalose-6-phosphate synthase/phosphatase pair**. An exhaustive proteome search returned glgA, glgB, glgE, glgP, and glgX but no glgC and no OtsAB. This configuration — the complete GlgE-pathway gene set present while glgC is absent — parallels *Streptomyces venezuelae* and supports the inference that the **maltose-1-phosphate/GlgE route is the primary (and possibly sole) pathway to intracellular branched α -glucan** in this organism. Because OtsAB is absent, the trehalose pool feeding module step 1 most plausibly derives from the **TreYZ pathway** (treY/PP_4053 + treZ/PP_4051), which produces trehalose from maltooligosaccharides/glycogen, and/or from environmental trehalose.

For curation, the practical message is that the KEGG ppu00500 candidate list of 18 genes is **over-broad**: only **PP_4058, PP_4059, and PP_4060** are true members of this specific biosynthetic module. The remaining candidates belong to neighboring processes — cellulose (β -1,4-glucan) synthesis, the TreYZ trehalose-*from*-glycogen route, classic glycogen turnover, and central hexose-phosphate metabolism — and should be excluded from the module while remaining valid members of the broader starch/sucrose map. All module gene calls are **homology-grade (evidence PE=3)**; no direct KT2440 biochemistry exists, but same-species *P. putida* TreS biochemistry and UniProt's explicit pathway assignment for PP_4060 substantially strengthen the calls. **Bottom line**: mark the module **SATISFIED** with a three-gene membership {PP_4058, PP_4059, PP_4060}; flag glgA/PP_4050 as *candidate_uncertain* given the GlgC gap; and exclude the remaining 14 candidates from this module.

2. Target-Organism Pathway Definition

2.1 What the module includes

The TreS–Mak–GlgE–GlgB module is a **reusable, cytosolic bacterial pathway that converts trehalose into a branched α -1,4/ α -1,6-glucan** through four sequential reaction roles:

1. **Trehalose $\rightarrow$ maltose** — trehalose synthase (TreS) operating in the maltose-forming (catabolic) direction; maltose α -D-glucosyltransferase activity (EC 5.4.99.16).
2. **Maltose $\rightarrow$ α -maltose 1-phosphate** — maltokinase (Mak/Pep2); carbohydrate kinase activity (EC 2.7.1.175).
3. **α -maltose 1-phosphate + [α -1,4-glucan]_n $\rightarrow$ [α -1,4-glucan]_{n+2} + P_i** — GlgE maltosyltransferase; extends chains by maltosyl units (EC 2.4.99.16).
4. **α -1,4-glucan $\rightarrow$ branched α -glucan** — GlgB branching enzyme introduces α -1,6 linkages (EC 2.4.1.18).

The canonical four-step scheme was defined in *Mycobacterium tuberculosis* by Kalscheuer et al. (PMID: 20305657), who described "a new pathway from trehalose to alpha-glucan ... comprising four enzymatic steps mediated by TreS, Pep2, GlgE ... and GlgB."

2.2 Neighboring pathways to keep separate

For accurate module boundaries in KT2440, the following processes — although co-listed under KEGG ppu00500 — must be **excluded**:

- **Cellulose (β -1,4-glucan) synthesis:** bcsA/PP_2635 (EC 2.4.1.12, GT2/PilZ). This makes a β -linked, not α -linked, glucan and is mechanistically unrelated.
- **TreYZ trehalose-*from*-glycogen pathway:** treY/PP_4053 (EC 5.4.99.15) + treZ/PP_4051 (EC 3.2.1.141). This runs in the *reverse* metabolic direction (glucan $\rightarrow$ trehalose) and is a *supplier* of trehalose, not part of the α -glucan biosynthetic module itself.
- **Classic glycogen/maltodextrin turnover:** glgA/PP_4050 (EC 2.4.1.21), glgP/PP_5041 (EC 2.4.1.1), glgX/PP_4055 (glycogen debranching), malQ/PP_4052 (amylomaltase, GH77).
- **Central hexose-phosphate metabolism:** glk, pgi1/pgi2, pgm, algC, cpsG, galU.

2.3 Alternate names and database definitions

- The module is variously called the "**GlgE pathway**", the "**TreS-Mak-GlgE pathway**", or the "**trehalose-to- α -glucan**" pathway.
- The maltokinase is called **Mak** in the generic module but **Pep2** in mycobacterial literature (e.g., Rv0127).
- KEGG folds all of these genes into the broad overview map **ppu00500 "Starch and sucrose metabolism"**, which is far broader than the biosynthetic module and is the source of the candidate-list over-breadth.
- GlgE is EC 2.4.99.16; its systematic name is **(1 $\rightarrow$ 4)- α -D-glucan: α -maltose-1-phosphate α -D-maltosyltransferase (GMPMT).**

3. Expected Step Model and Satisfiability

The table below maps each expected step to its KT2440 gene, copy number, evidence type, and disposition.

Step	Reaction role	EC	KT2440 gene	UniProt	Copy number	Disposition
1	Trehalose $\rightleftharpoons$ maltose (TreS)	5.4.99.16	PP_4059 (treSB, fused) + PP_2918 (treSA, standalone)	Q88FN0 / Q88IT1	2 paralogs	covered
2	Maltose $\rightarrow$ α -maltose 1-P (Mak)	2.7.1.175	PP_4059 (treSB, fused C-terminal domain)	Q88FN0	1	covered
3	α -glucan extension (GlgE)	2.4.99.16	PP_4060 (glgE)	Q88FM9	1	covered
4	α -1,6 branching (GlgB)	2.4.1.18	PP_4058 (glgB)	Q88FN1	1	covered

Satisfiability verdict: all four steps covered. Steps 2, 3, and 4 are single-copy and unambiguous. Only step 1 (TreS) has paralog redundancy (see §4.5). No expected step is missing, and none is judged *not_expected_in_target_taxon*.

```

[trehalose]
|
| Step 1  TreS (EC 5.4.99.16)  -- PP_4059 N-terminal
v          treSB fusion          GH13 domain
[D-maltose]
|
| Step 2  Mak (EC 2.7.1.175)   -- PP_4059 C-terminal
v          treSB fusion          kinase-like domain
[alpha-maltose 1-phosphate]
|
| Step 3  GlgE (EC 2.4.99.16)  -- PP_4060
v          maltosyltransferase
[alpha-1,4-glucan]n+2 (chain extended by maltosyl units)
|
| Step 4  GlgB (EC 2.4.1.18)   -- PP_4058
v          branching enzyme
[branched alpha-1,4/1,6-glucan] <-- final product

NOTE: GlgC (ADP-glucose pyrophosphorylase, EC 2.7.7.27) ABSENT
      OtsAB (EC 2.4.1.15 / 3.1.3.12) ABSENT
      => GlgE/maltose-1-P route is the PRIMARY alpha-glucan pathway

```

(from TreYZ: PP_4053 + PP_4051, or environment)

4. Candidate Genes and Evidence

4.1 The three true module members (Findings F001, F006, F007)

All four module steps are encoded within the contiguous PP_4050–PP_4060 glycogen/trehalose locus. UniProt/InterPro annotation (homology-based, PE=3) maps every step:

- **PP_4058 / GlgB / Q88FN1** — 1,4- α -glucan branching enzyme (EC 2.4.1.18), GH13 GlgB subfamily (InterPro IPR006407, GlgB-specific). Single-copy for EC 2.4.1.18. Covers **step 4**.
- **PP_4059 / TreSB / Q88FN0** — bifunctional TreS–maltokinase fusion (see §4.2). Covers **steps 1 and 2**.
- **PP_4060 / GlgE / Q88FM9** — maltosyltransferase (EC 2.4.99.16), GH13 GlgE subfamily (InterPro IPR026585, GlgE-specific). Single-copy for EC 2.4.99.16. Covers **step 3**.

UniProt's curated FUNCTION comment for PP_4060 is decisive for module assignment: GlgE is "a maltosyltransferase that uses maltose 1-phosphate (M1P) ... involved in a branched alpha-glucan biosynthetic pathway from trehalose, together with TreS, Mak and GlgB." This is an explicit, curator-level placement of PP_4060 into exactly this module (Finding F006).

4.2 PP_4059 — the TreS–Mak fusion (Finding F002)

PP_4059 (Q88FN0, 1106 aa) is the pivotal gene. It carries **two EC numbers on one polypeptide**: 5.4.99.16 (maltose α -D-glucosyltransferase = trehalose synthase) and 2.7.1.175 (maltokinase). Its domain architecture is an N-terminal GH13 α -amylase/TreS module (InterPro IPR012810 TreS/ α -amylase_N + IPR006047 GH13; Pfam PF00128) fused to a C-terminal aminoglycoside-phosphotransferase-like maltokinase (InterPro IPR012811 TreS_maltokinase_C + IPR040999 Mak_N_cap + IPR011009 kinase-like superfamily; Pfam PF18085 Mak_N_cap + PF16657).

This fusion contrasts with mycobacteria, where TreS (Rv0126) and Pep2/Mak (Rv0127) are **separate genes**. Roldán et al. (PMID: 23901909) showed that "Together with Pep2, TreS forms a hetero-octameric complex, and we demonstrate that complex formation markedly accelerates maltokinase activity of Pep2." The KT2440 fusion co-localizes the two activities covalently rather than by complex formation — an architectural variant that does not change the reaction roles. This same physical association is structurally corroborated by the crystal structure of the TreS:Pep2 complex (PMID: 30877199).

4.3 Same-species biochemical support (Finding F006)

Although all KT2440 module calls are homology-grade (PE=3), two *Pseudomonas*-genus biochemical studies materially strengthen step 1:

- **TreS from *P. putida* P06** experimentally catalyzes the reversible maltose $\rightleftharpoons$ trehalose interconversion (PMID: 24563286): "Trehalose synthase (TreS) from *Pseudomonas putida* P06 catalyzes the reversible interconversion of maltose and trehalose." This is direct, *same-species* biochemistry for the TreS activity.
- A ~1,122-aa *Pseudomonas* sp. P8005 TreS (PMID: 23715900) closely matches the 1106-aa length of KT2440 treSB, corroborating that the extended/fused architecture is characteristic within the genus.

4.4 The GlgC/OtsAB gap and pathway primacy (Finding F004)

An exhaustive UniProt proteome search (organism_id:160488) returned glgA, glgB, glgE, glgP, glgX but **no glgC** — EC 2.7.7.27 (glucose-1-phosphate adenylyltransferase) is absent, and no ADP-glucose-producing enzyme appears among any annotated adenylyltransferase. **OtsAB is also absent**: EC 2.4.1.15 (trehalose-6-P synthase) and EC 3.1.3.12 (trehalose-6-P phosphatase) returned zero hits.

This matters for pathway primacy. In *M. tuberculosis*, α -glucan is "exclusively assembled intracellularly utilizing the building block α -maltose-1-phosphate as the substrate for the maltosyltransferase GlgE, with subsequent branching ... by the branching enzyme GlgB" (PMID: 27513637). In *S. venezuelae*, "The genome ... contains all the genes coding for the GlgE pathway enzymes but none of those of related pathways, including glgC and glgA of the glycogen pathway" (PMID: 27121970). KT2440's configuration (complete GlgE gene set, glgC absent) parallels these precedents and supports the inference that the **GlgE/maltose-1-P route is the primary intracellular α -glucan pathway** in KT2440.

4.5 Paralog ambiguity and trehalose supply (Finding F005)

Two TreS paralogs exist in KT2440: - **treSB / PP_4059 / Q88FN0** — 1106 aa, **fused** to the maltokinase domain, in the glg/tre cluster. - **treSA / PP_2918 / Q88IT1** — 688 aa, **standalone** (no Mak domain), at a distant locus.

Only PP_4059 carries the maltokinase activity required for step 2, so **PP_4059 is the module-relevant TreS**; PP_2918 is a paralog that may contribute maltose but cannot phosphorylate it. Because OtsAB is absent, the trehalose pool feeding step 1 most plausibly derives from **TreYZ** (treY/PP_4053 EC 5.4.99.15 + treZ/PP_4051 EC 3.2.1.141), which converts maltooligosaccharides/glycogen into trehalose, and/or from environmental trehalose.

4.6 Full 18-gene candidate disposition (Finding F003)

Gene	Locus	UniProt	Annotation / EC	Process	Module disposition
glgB	PP_4058	Q88FN1	Branching enzyme (2.4.1.18)	α -glucan branching	MODULE — step 4
treSB	PP_4059	Q88FN0	TreS + Mak fusion (5.4.99.16; 2.7.1.175)	trehalose→maltose→M1P	MODULE — steps 1+2
glgE	PP_4060	Q88FM9	Maltosyltransferase (2.4.99.16)	α -glucan extension	MODULE — step 3
glgA	PP_4050	Q88FN9	Glycogen synthase (2.4.1.21)	glycogen synth (ADP-glc)	candidate_uncertain
treZ	PP_4051	Q88FN8	Trehalohydrolase (3.2.1.141)	TreYZ (trehalose supply)	exclude (supplier)
malQ	PP_4052	Q88FN7	Amylomaltase (2.4.1.25)	maltodextrin turnover	exclude
treY	PP_4053	Q88FN6	Maltooligosyl-tre synthase (5.4.99.15)	TreYZ (trehalose supply)	exclude (supplier)
glgX	PP_4055	Q88FN4	Debranching (3.2.1.33)	glycogen turnover	exclude
glgP	PP_5041	Q88CY8	Glucan phosphorylase (2.4.1.1)	glycogen turnover	exclude
bcsA	PP_2635	Q88JL4	Cellulose synthase (2.4.1.12)	β -1,4-glucan (cellulose)	exclude (β -glucan)
glk	PP_1011	Q88P42	Glucokinase (2.7.1.2)	central metabolism	exclude
bglX	PP_1403	Q88N13	β -glucosidase (3.2.1.21)	glycoside hydrolysis	exclude
cpsG	PP_1777	Q88LZ9	Phosphomannomutase (5.4.2.8)	central metabolism	exclude
pgi1	PP_1808	Q88LW9	G6P isomerase (5.3.1.9)	central metabolism	exclude
pgm	PP_3578	Q88GY7	Phosphoglucomutase (5.4.2.2)	central metabolism	exclude
galU	PP_3821	Q88GA4	UDP-glucose PPase (2.7.7.9)	UDP-glucose supply	exclude

Gene	Locus	UniProt	Annotation / EC	Process	Module disposition
pgi2	PP_4701	Q88DW7	G6P isomerase (5.3.1.9)	central metabolism	exclude
algC	PP_5288	Q88C93	PMM/PGM (5.4.2.2; 5.4.2.8)	central metabolism	exclude

5. Gaps, Ambiguities, and Likely Over-Annotations

- 1. KEGG ppu00500 breadth is the main over-annotation risk.** 15 of 18 candidates are not members of the biosynthetic module. Curators pulling from the ppu00500 bucket will over-populate the module unless it is trimmed to {PP_4058, PP_4059, PP_4060}. This is a bucket-scope artifact, not an error in any individual gene annotation.
- 2. glgA/PP_4050 is genuinely ambiguous (candidate_uncertain).** Classic GlgA glycogen synthase (EC 2.4.1.21) uses ADP-glucose produced by GlgC. With **glgC absent**, the physiological substrate/role of PP_4050 is unclear — it may be non-functional for its annotated reaction, may use a different sugar-nucleotide donor, or may be an over-propagated annotation. This gene warrants explicit *candidate_uncertain* flagging and full review.
- 3. TreS paralog ambiguity.** treSA/PP_2918 and treSB/PP_4059 both carry EC 5.4.99.16. Only the fused PP_4059 is module-relevant. A naive EC-based mapping could incorrectly assign PP_2918 to the module.
- 4. All calls are homology-grade (PE=3).** No direct KT2440 experiment demonstrates GlgE maltosyltransferase activity, maltokinase activity, or α -glucan production. A dedicated PubMed search for KT2440-specific GlgE/ α -glucan/maltose-1-phosphate evidence returned **zero papers** (Finding F007). Confidence rests on: (a) same-genus *P. putida* TreS biochemistry; (b) UniProt's explicit pathway-level curation of PP_4060; and (c) InterPro subfamily-specific signatures (GlgE-specific IPR026585, GlgB-specific IPR006407) that are diagnostic rather than broad.
- 5. Broad EC/GO mappings.** GlgE's EC (2.4.99.16) and GH13 membership are specific enough, but generic "hexosyltransferase" GO terms could over-generalize. GlgB shares GH13 with many α -amylase-family enzymes; rely on the GlgB-specific InterPro signature, not GH13 alone.

6. Module and GO-Curation Recommendations

Module step	Recommended status	Gene(s)	Rationale
1 — TreS (trehalose→maltose)	covered	PP_4059 (module), PP_2918 (paralog)	Fused GH13 TreS domain; same-species <i>P. putida</i> TreS biochemistry
2 — Mak (maltose→M1P)	covered	PP_4059 (fused kinase domain)	Single-copy EC 2.7.1.175; Mak_N_cap + kinase-like domains
3 — GlgE (α -glucan extension)	covered	PP_4060	Single-copy; GlgE-specific InterPro; explicit UniProt pathway comment
4 — GlgB (α -1,6 branching)	covered	PP_4058	Single-copy; GlgB-specific InterPro

Module verdict: SATISFIED.

Additional curation actions: - **Trim module membership** to exactly {PP_4058, PP_4059, PP_4060}. Do not inherit the full ppu00500 candidate list. - **Flag glgA/PP_4050 as candidate_uncertain** and note the GlgC gap in the module rationale. - **Document the fusion architecture** of PP_4059 explicitly (steps 1+2 in one protein) so that automated "one gene per step" checks do not falsely report step 2 as a gap. - **Record the GlgC/OtsAB absence** as supporting evidence that the GlgE route is primary; this is a positive discriminating feature, not a gap in *this* module. - **Note trehalose supply via TreYZ** (PP_4053/PP_4051) as an upstream dependency, but keep TreYZ genes *out* of the module itself (they are suppliers running in the reverse direction). - **GO terms:** existing GO/EC mappings are adequate; no new GO term request appears necessary. If a module-specific GO annotation is desired, the " α -glucan biosynthetic process from trehalose" framing is well supported by the mycobacterial literature and can be transferred with a homology (ISS/IEA) evidence code, explicitly flagged as inferred. - **Module boundaries are correct for this organism;** the generic module does not need revision for KT2440 beyond documenting the TreS–Mak fusion, which the generic scope already anticipates ("TreS and Mak may be separate proteins or domains of one fusion protein").

7. Genes to Promote to Full fetch-gene Review

Priority order:

1. **PP_4059 (treSB, Q88FN0)** — highest priority. Confirm the bifunctional TreS+Mak assignment, both EC numbers, domain boundaries, and that this (not PP_2918) is the module-relevant TreS. The dual-EC fusion is the single most curation-sensitive call in the module.
 2. **PP_4060 (glgE, Q88FM9)** — confirm GlgE-specific subfamily assignment and the maltose-1-phosphate substrate; UniProt already places it in the pathway, but a full review cements step 3.
 3. **PP_4050 (glgA, Q88FN9)** — resolve the candidate_uncertain status: is glycogen synthase functional/relevant given the GlgC gap? Determine sugar-nucleotide donor availability.
 4. **PP_4058 (glgB, Q88FN1)** — confirm branching-enzyme assignment (lower priority; annotation is clean and single-copy).
 5. **PP_2918 (treSA, Q88IT1)** — clarify the role of the standalone TreS paralog and whether it feeds maltose into the pathway independently of PP_4059.
-

8. Evidence Base and Key References

PMID	Title (abbrev.)	Organism	How it supports the review
20305657	Self-poisoning of <i>M. tuberculosis</i> by targeting GlgE	<i>M. tuberculosis</i>	Defines the canonical four-step TreS–Pep2(Mak)–GlgE–GlgB module mapped onto KT2440 (F001)
23901909	α -glucan synthesis via a hetero-octameric TreS–Pep2 complex	<i>M. tuberculosis</i>	Shows TreS and Mak are separate, complex-forming proteins in mycobacteria; KT2440 fuses them into PP_4059 (F002)
27513637	Metabolic network for α -glucan biosynthesis	<i>M. tuberculosis</i>	α -glucan "exclusively assembled ... from α -maltose-1-phosphate ... GlgE ... GlgB"; supports GlgE route as primary where glgC-independent (F004)
27121970	glgE null mutant accumulates α -maltose 1-phosphate	<i>S. venezuelae</i>	Precedent for a complete GlgE pathway with glgC/glgA absent, paralleling KT2440 (F004)
24563286	Homology modeling and function of TreS from <i>P. putida</i> P06	<i>P. putida</i>	Direct same-species biochemistry: TreS catalyzes reversible maltose $\rightleftharpoons$ trehalose, supporting step 1 (F006)
23715900	Novel TreS from marine <i>Pseudomonas</i> sp. P8005	<i>Pseudomonas</i> sp.	~1,122-aa TreS matches the fused KT2440 treSB length (1106 aa), corroborating fusion architecture in the genus (F006)
20118231	GMPMT transfers maltose from maltose-1-P to glycogen	<i>M. smegmatis</i>	Biochemically characterizes the GlgE (GMPMT) reaction and proposes trehalose $\rightarrow$ glycogen via TreS+Mak+GMPMT
26616850	Structure of <i>M. thermoresistibile</i> GlgE	<i>M. thermoresistibile</i>	Structural context for GlgE catalytic mechanism (step 3)
27531751	Acceptor/secondary sites of <i>S. coelicolor</i> GlgE	<i>S. coelicolor</i>	Defines GlgE acceptor specificity, informing step-3 substrate model
26245983		<i>M. tuberculosis</i>	

PMID	Title (abbrev.)	Organism	How it supports the review
	Crystal structures of <i>M. tuberculosis</i> GlgE + inhibitors		GlgE elongates cytosolic branched α -glucan using M1P; structural validation
30877199	Crystal structure of the TreS:Pep2 complex	<i>M. tuberculosis</i>	Structural basis of the TreS–Mak interaction that KT2440 replaces with a fusion
25137149	Poly-hydroxypyrolidine GlgE inhibitor	<i>M. tuberculosis</i>	Confirms GlgE M1P substrate and the toxic-M1P-accumulation phenotype on GlgE loss
24820953	Homology model of <i>M. tuberculosis</i> GlgE	<i>M. tuberculosis</i>	GlgE belongs to the α -amylase (GH13) family; five-domain architecture

Evidence provenance summary: - **Direct for target species (*P. putida*):** TreS maltose $\rightleftharpoons$ trehalose biochemistry (PMID: [24563286](#)) and genus-level TreS length (PMID: [23715900](#)). These are strongest for step 1. - **Direct for target strain (KT2440):** None. All gene calls are PE=3 homology inferences plus UniProt/InterPro curation. - **Transferred from related organisms (mycobacteria, *Streptomyces*):** The four-step module scheme, GlgE mechanism, and GlgC-independence precedent. Transfer strength is **moderate-to-strong** for the reaction roles (deeply conserved GH13 chemistry) but **weaker** for regulation, flux direction, and the physiological role of the α -glucan product in KT2440.

9. Limitations and Knowledge Gaps

- 1. No strain-specific experiments.** No published study demonstrates GlgE-pathway function, α -glucan production, or maltose-1-phosphate metabolism directly in KT2440. All module calls are homology-grade (PE=3).
- 2. glgA/PP_4050 role unresolved.** The GlgC gap leaves the classic glycogen-synthase branch physiologically ambiguous; whether glycogen and GlgE-derived α -glucan are the same or distinct polymers in KT2440 is unknown.
- 3. Trehalose source inferred, not measured.** TreYZ is the most plausible trehalose supplier given the OtsAB absence, but flux from TreYZ (or environmental uptake) into module step 1 has not been quantified in KT2440.

4. **Product identity and localization unknown.** Whether the α -glucan is intracellular storage, a capsule, or both — as it varies across mycobacteria — is not established for KT2440.
 5. **Directionality of TreS.** TreS is reversible; whether PP_4059 runs net in the maltose-forming direction in vivo (as the module requires) versus the trehalose-forming direction depends on cellular conditions not measured here.
 6. **Absence calls rely on annotation completeness.** The GlgC/OtsAB "absent" conclusions rest on UniProt proteome annotation; a highly diverged or mis-annotated ortholog cannot be fully excluded without dedicated HMM/structural searches.
-

10. Proposed Follow-up Experiments and Actions

Curation actions (immediate): 1. Set module status to **SATISFIED** with membership {PP_4058, PP_4059, PP_4060}; record the PP_4059 fusion so automated step-2 gap checks are suppressed. 2. Flag **glgA/PP_4050 as candidate_uncertain**; promote PP_4059, PP_4060, and PP_4050 to full `fetch-gene` review (§7). 3. Annotate the **GlgC/OtsAB absence** as a positive discriminator that the GlgE route is primary; note **TreYZ (PP_4053/PP_4051)** as the upstream trehalose supplier, kept outside the module.

Bioinformatic follow-up: 4. Run HMM/structure-based searches (e.g., against Pfam GlgC HMMs and AlphaFold models) to confirm GlgC and OtsAB are truly absent rather than diverged. 5. Verify PP_4059 domain boundaries and confirm that maltokinase catalytic residues are intact in the C-terminal domain.

Experimental validation (if warranted): 6. Construct Δ glgE (PP_4060) and Δ treSB (PP_4059) mutants and assay for α -maltose-1-phosphate accumulation and loss of α -glucan — the diagnostic phenotype seen in *S. venezuelae* (PMID: 27121970) and the toxic-accumulation death phenotype in mycobacteria (PMID: 25137149). 7. Assay recombinant PP_4059 for both TreS and maltokinase activities to confirm the fusion is doubly active, and PP_4060 for maltosyltransferase activity using α -maltose-1-phosphate as donor. 8. Characterize the α -glucan product (linkage analysis, localization) to establish its physiological role in KT2440. 9. Test whether TreYZ supplies the trehalose feeding step 1 via Δ treYZ metabolite profiling.

Prepared as a species-aware module satisfiability review. All gene-level calls for KT2440 are homology-grade (PE=3) unless otherwise noted; species transfer from mycobacteria/Streptomyces is explicitly flagged throughout.

Generated by OpenScientist — Scientific Hypothesis Agent for Novel Discovery